

BINF6999 Final Reflection: When Strong Evidence Tests as Noise

Project: A Mechanism-Informed Gene Signature for Regenerative vs. Fibrotic Tendon Healing

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What

The second half of the project was the integration phase: combining the pySCENIC regulon analysis, the CellRank fate trajectories, and the LIANA ligand-receptor analysis into one mechanistic circuit, then compressing that circuit into a gene panel and a continuous Healing Index. By mid-July I had a story I was proud of, and two results sat at its center. The first was $Efn1 \rightarrow Eph3$, which my notes described as the strongest differential niche signal between the innervated and denervated conditions, based on its `delta_rank`, the difference between a ligand-receptor pair's aggregate rank in the two conditions. The second was a feedback loop: six of the fifteen cross-dataset T-FAP regulons appeared to target `Pdgfra`, which I had framed as the fibrotic program reshaping the cell's own receptivity to niche signals. Both were drawn into the circuit figure and headed for the poster and report.

While drafting the methods section, I had to write down, for each claim, the statistical test that supported it. For these two, there was nothing to write. LIANA had run with permutation testing switched off, so `delta_rank` carried no significance test of any kind. The feedback loop was a raw overlap count between gene lists of very unequal sizes, never compared against chance. In mid-July I re-ran LIANA with 100 permutations and tested the `Pdgfra` overlap with per-dataset hypergeometric enrichment, Fisher-combined and BH-corrected across the fifteen regulons. $Efn1 \rightarrow Eph3$ returned $p = 1.0$. The feedback loop produced nothing significant even at an uncorrected threshold; the best case, `Fos`, reached $p \approx 0.056$. I retracted both claims and regenerated the circuit figure with those edges removed.

That experience turned into a pre-specified stress-testing protocol for all four headline findings. The zero-TSPC-regulon result from the first half survived a 200-iteration power-matched stability selection. The CellRank result held only in part: the absolute fate probabilities shifted 5 to 7 points under an alternate pseudotime root, though the directional shift and 94% of the 119-gene candidate list were stable. The other two findings, as above, were retracted.

So What

The question that stayed with me was why the two false results had felt like the strongest ones. They were vivid, mechanistic, and story-shaped; the feedback loop in particular was the most publishable-sounding sentence in the project. Both came from procedures that produce compelling-looking output without testing it: a difference between two independently computed rank aggregates, and a set overlap. When I

checked systematically, only 8 of the top 20 innervated-favoured pairs and 12 of the top 20 denervated-favoured pairs by Δ _rank cleared $p < 0.05$. On this data, the metric I had been treating as evidence identified reproducible signals less than two-thirds of the time.

The uncomfortable part was recognizing the asymmetry in my own behaviour. My intermediate reflection described the opposite case: an unwelcome null, the zero TSPC regulons, that I initially treated as a bug and interrogated for days until I accepted it was real. This half exposed the mirror image. I had spent days interrogating the zero because I disliked it, and zero minutes interrogating the feedback loop because I liked it. Skepticism applied only to unwelcome results is not skepticism; it is negotiation with the data. Modern single-cell tools make this failure mode easy, because the pipeline always returns something. GRNBoost2 will always produce a network, LIANA will always produce a ranking, and none of these tools refuses to run when the input question is ill-posed. The default status of an untested pattern is noise, and the person running the pipeline is the only quality gate.

What convinced me the stress-testing was worth its cost came at report-writing time. Specifying each test before knowing its outcome, the same logic as pre-registration, is what let me present the surviving claims with confidence. The report's central finding is an inversion of my original hypothesis: fibrosis appears to be the default program, and innervation the constraint holding it back. That inversion rests on the zero-TSPC result, and it is only credible because that result passed through the same gauntlet that broke two others. Retracting my own results three weeks before the deadline felt like losing ground. Writing the report, the stress-test register turned out to be the section I could defend most confidently.

Now What

I used to think validation was something you applied to results that looked wrong, and that a strong result earned trust by fitting the biology. Now I understand that the results that fit best are exactly the ones I am least equipped to doubt on my own, so doubt has to be structural rather than mood-driven: specified in advance, applied uniformly, and applied hardest to the claims I most want to keep.

Three concrete practices come out of this and carry into my future work. First, a claims register: every claim in my final report now travels with the test it survived, and in future projects no result graduates from exploratory to reported without a pre-specified test, recorded before I see whether the result passes. Second, a default classification: rank differences, overlap counts, and top-N lists are hypothesis generators until tested, however coherent the biology they suggest. Third, a scheduling change: adversarial re-analysis is now a budgeted pipeline stage, the "b" notebooks in my repository, rather than an epilogue squeezed in if time allows. I also learned that writing the methods section is itself a validation tool. The missing tests were exposed not by re-examining figures but by the simple obligation to complete the sentence "significance was assessed by". I will draft methods while analyzing, not after, because the sentences I cannot finish are the claims I cannot support.

The first half of this project taught me that zero can be an answer. The second half taught me the complement: a compelling number is not yet evidence. What changed between June and August is where I place my doubt. It used to go toward the results that disappointed me. It now goes, first and hardest, toward the ones I want to be true.